

L-3. Landslide Treatment				
Degraded Soil & Land Improvement	Medium	High		
Measure Type	Cost	Preferences by Community	Hazards	Implementation Time of Measures

Descriptions									
Measures	The term "landslide treatment" describes a group of environmental and engineering methods intended to stabilize slopes and lessen or stop landslides from happening. Several factors, including heavy rainfall, earthquakes, volcanic eruptions, and human activities such as construction or deforestation, can cause landslides. Reducing the risk of landslides, protecting infrastructure, the environment, and human life, as well as reestablishing stability in the impacted area, are the objectives of landslide treatment.								
	The CHT in Bangladesh are particularly vulnerable to landslides due to its hilly topography, steep slopes, and intense monsoon rains. Numerous landslide treatment techniques are used to reduce the hazards and stabilize the slopes in the region, including terracing, retaining walls and reinforcement, slope reshaping and grading, geotechnical engineering solutions, and reforestation and vegetative cover. These techniques aim to preserve residential and agricultural areas, stabilize slopes, and prevent soil erosion.								
Location	R-114	R-99	R-65	K-98	K-76	K-71	B-173	B-77	B-76
	■	■	■	■	■	■	■	■	■
Applicable Sites	• Landslide-Prone Hillslopes (>15%) • Deforested and Degraded Slopes • Settlements & Roadside Slopes • Agricultural Terraces & Plantation Areas								
Slope	Less than 3%		3% to 15%		15% to 30%		30% to 60%		Above 60%
					√		√		
Design options and performance	• Terracing & Step Farming – Reduces surface runoff and soil displacement, improving water retention and crop productivity. • Retaining Walls & Reinforcement Structures – Uses concrete, gabions, or wooden walls to support unstable slopes. • Slope Reshaping & Grading – Reduces steepness, improving slope stability and drainage. • Geotechnical Solutions (Soil Nailing, Anchors, Pile Foundations) – Strengthens high-risk slopes using deep structural reinforcements. • Reforestation & Vegetative Cover – Uses deep-rooted trees, grasses, and vetiver grass barriers to bind soil and reduce erosion. • Drainage Systems (Surface & Subsurface) – Prevents water accumulation and reduces pore water pressure.								

Feasibility criteria	- To determine whether the area is vulnerable to erosion, slippage, or instability, the type of rock and soil present must be evaluated. It is necessary to perform thorough analyses of the slope's angle and current stability conditions. To determine the risk of water-induced landslides, it is important to examine the permeability, structural integrity, and groundwater levels of subsurface rock and soil. - Protecting the forest cover, water bodies, and wildlife is essential in the CHT, which are home to a wealth of biodiversity. Priority should be given to environmentally beneficial strategies like agroforestry and afforestation. - Future climate scenarios should be taken into account while designing the procedures. These scenarios should include changes in temperature, rainfall patterns, and extreme weather events that may have an impact on the occurrence of landslides.			
Operation and maintenance	To ensure the long-term efficacy of landslide treatment, the stabilization measures put in place must be regularly monitored, repaired, and managed. Educating the neighborhood to help with upkeep duties including terrace cleaning, replanting, and reporting any indications of unstable soil.			
Cost	Medium			
Benefits				
	Type	Low	Medium	High
	Canopy Cover Increase			
	Carbon Sequestration			
	Crop Production Increase			
	Crop Failure Risk Reduction			
	Ensuring Food Security			
	Fruit Production Increase			
	Forestry Goods Production Increase			
	Livestock Production Increase			
	Fisheries Production Increase			
	Alternative Livelihood Generation			
	Increase of Income			
	Soil Erosion Control			
	Improved Soil Health			
	Soil Fertility Increase			
	Sustainable Water Supply			
	Water Quality Improvement			
	Water Purification			
	River Bank Erosion Control			
	Surface Runoff Reduction			
	Groundwater Recharge			
	Pest and Disease Management			
	Reduction of Pesticides and Fertilizer Use			

	Ecosystem and Biodiversity Increase			
	Better Communications			
	Better Education			
	Better WASH facilities			
Implementation Challenges				
	Type	Low	Medium	High
	Land unavailable/high land area required			
	Supply chain disconnection			
	Land right related problems			
	High human investment capital need			
	Insufficient infrastructures			
	Less demand of agroforestry product			
	Environmentally unsustainable			
	Socially unacceptable			
	Less beneficial			
Environmental performance	- For the purpose to stabilize slopes and prevent soil erosion, landslide treatment techniques like terracing, growing vegetation, and retaining walls are implemented. The region's agriculture depends on the fertile soil, which is maintained in this way. - The restoration of natural ecosystems by forestry and reforestation as part of landslide treatment enhances the region's biodiversity. Local ecosystems, wildlife, and the larger food chain are all supported by native trees and plants. - The region is more sensitive to extreme weather events, like heavy monsoon rains, which are predicted to grow more frequent and intense owing to climate change, when slopes are stabilized and water flow is managed by drainage systems.			
Remarks	This measure will aid landscape restoration in CHT of Bangladesh along with creating alternative livelihood opportunities.			